26900408 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEECT310 - ROBOTICS AND INDUSTRIAL AUTOMATION

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. What are the main objectives of using robots in industrial applications? [U] / [CO1]
2. What is a geared DC motor? [U] / [CO1]
3. What is meant by position control in a servo motor? [U] / [CO1]
4. List any two important features of Industry 4.0. [U] / [CO1]
5. What is a transistor output module? [K] / [CO2]
6. What is a PLC input? [K] / [CO2]
7. What is an alarm output in a sensor-based PLC system? [K] / [CO2]
8. What is the role of an output status indicator? [U] / [CO2]
9. What is a coil in Ladder Logic? [K] / [CO2]

10. What is interlocking in motor control? [U] / [CO2]

11. List any two applications of control systems in robotics. [K] / [CO3]

12. State one advantage of robotic spot welding. [U] / [CO3]

13. What is a pick-and-place robot? [U] / [CO3]

14. What is the role of a controller in an automatic water level system? [K] / [CO3]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

15. Explain the important features of a stepper motor that make it suitable for position control applications. [U] / [CO1]

16. Explain the basic working principle of a piezoelectric sensor and mention its applications. [U] / [CO1]

17. Explain the role of electrical actuators in robotic and automated systems. [U] / [CO1]

18. Design and explain a start-stop latching circuit for a motor using Ladder Logic. [A] / [CO2]

19. Explain how a PLC can monitor gas concentration and generate an alarm when a preset limit is exceeded. [A] / [CO2]

20. Explain the basic working principles of temperature sensor and its PLC interfacing requirements. [U] / [CO2]

21. Explain the major components of an automatic conveyor belt mechanism and state the function of each. [A] / [CO3]

22. Explain the importance of an automatic conveyor belt mechanism in industrial material handling. [U] / [CO3]

[5x6 = 30]